A six-node counterexample to generic local implying generic global network identifiability

Research note prepared with Codex

8 October 2026

Abstract

We construct a directed network with six nodes, nine independent unknown edge transfer functions, three excited nodes and three measured nodes. Its observation map $F(G) = C(I - G)^{-1}B$ has generic fiber cardinality two and a generically nonsingular Jacobian. Thus generic local identifiability does not imply generic global identifiability under the well-posedness assumptions in Legat and Hendrickx (2020). No unknown input or measurement matrices, repeated parameter constraints, known nonzero edges, or self-loops are used. A stronger admissibility condition $\rho(G) < 1$, introduced in later work, excludes the alternative branch in this example; the result does not refute that more restricted formulation.

Model and scope. The original question concerns $w = Gw + Br$, $y = Cw$, with fixed node selection matrices B, C and independently unknown entries of G on a prescribed graph. The 2020 paper assumes that $I - G$ is invertible and studies the map at a fixed complex frequency [1]. In the notation of the supplied paper, $P = I - G$, $Q = B$, $R = C$, $S = 0$. All generic statements below mean outside a proper algebraic subset of the nine-dimensional parameter space. The same rational identities hold over $\mathbb{C}(q)$.

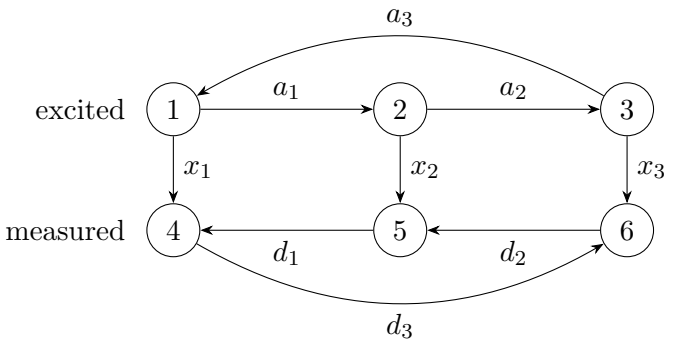
Set

$$A = \begin{pmatrix} 0 & 0 & a_3 \\ a_1 & 0 & 0 \\ 0 & a_2 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} 0 & d_1 & 0 \\ 0 & 0 & d_2 \\ d_3 & 0 & 0 \end{pmatrix}, \qquad X = \operatorname{diag}(x_1, x_2, x_3),$$

and define

$$G = \begin{pmatrix} A & 0 \\ X & D \end{pmatrix}, \qquad B = \begin{pmatrix} I_3 \\ 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & I_3 \end{pmatrix}. \tag{1}$$

The nine symbols a_i, d_i, x_i are independent unknowns. Nodes $1, 2, 3$ are excited and nodes $4, 5, 6$ are measured. The two directed cycles run in opposite directions, and three edges join corresponding nodes from the excited cycle to the measured cycle.



Write $\alpha = a_1a_2a_3$, $\delta = d_1d_2d_3$. Then

$$\det(I_6 - G) = (1 - \alpha)(1 - \delta), \qquad F = (I_3 - D)^{-1}X(I_3 - A)^{-1}.$$

On the generic set $x_1x_2x_3 \neq 0$, F is invertible. Its inverse is the particularly simple matrix

$$K = F^{-1} = (I_3 - A)X^{-1}(I_3 - D) = \begin{pmatrix} x_1^{-1} + a_3d_3x_3^{-1} & -d_1x_1^{-1} & -a_3x_3^{-1} \\ -a_1x_1^{-1} & x_2^{-1} + a_1d_1x_1^{-1} & -d_2x_2^{-1} \\ -d_3x_3^{-1} & -a_2x_2^{-1} & x_3^{-1} + a_2d_2x_2^{-1} \end{pmatrix}. \quad (2)$$

Proposition 1. *The network structure (1) is generically locally identifiable and generically two-to-one under $G \mapsto C(I_6 - G)^{-1}B$.*

Proof. Introduce the invertible rational change of coordinates

$$u_i = a_i/x_i, \qquad v_i = d_i/x_i, \qquad z_i = 1/x_i.$$

The six off-diagonal entries of K determine u_i, v_i directly. Its diagonal entries $e_i = K_{ii}$ satisfy, with indices modulo three,

$$e_i = z_i + \frac{u_{i-1}v_{i-1}}{z_{i-1}}. \quad (3)$$

For fixed u, v, the Jacobian with respect to z is

$$J_z = \begin{pmatrix} 1 & 0 & -u_3v_3/z_3^2 \\ -u_1v_1/z_1^2 & 1 & 0 \\ 0 & -u_2v_2/z_2^2 & 1 \end{pmatrix}, \qquad \det J_z = 1 - \frac{\prod_i u_iv_i}{\prod_i z_i^2} = 1 - \alpha\delta.$$

Thus the full nine-dimensional observation map has Jacobian rank nine generically. Inverting F and changing parameter coordinates preserve this rank on their domains. This proves generic local identifiability.

To prove the fiber cardinality, set $b_i = u_iv_i$. From (3),

$$z_2 = e_2 - \frac{b_1}{z_1}, \qquad z_3 = e_3 - \frac{b_2}{z_2}, \qquad z_1 = e_1 - \frac{b_3}{z_3}.$$

The last equation is a fixed-point equation for a composition of three Möbius transformations. Clearing its denominators gives

$$\begin{aligned} 0 = {} & (b_2 - e_2e_3)z_1^2 \\ & + (b_1e_3 - b_2e_1 - b_3e_2 + e_1e_2e_3)z_1 + b_1b_3 - b_1e_1e_3. \end{aligned} \quad (4)$$

Generically this is a nonzero quadratic, so there are at most two admissible solutions; z_2, z_3 and then a, d, x are determined by z_1. Zero values of x_i cannot give additional solutions, because the observed F is invertible.

We now explicitly exhibit the second solution and check admissibility. Define cyclically

$$k_i = a_id_i\frac{x_{i+1}}{x_i}, \qquad h_i = 1 + k_i + k_ik_{i+1}, \qquad r_i = \frac{k_ih_{i+1}}{h_i}, \qquad (a_i', d_i', x_i') = \frac{1}{r_i}(a_i, d_i, x_i). \quad (5)$$

These rational expressions are defined and nonzero generically. Let $L = k_1k_2k_3 = \alpha\delta$. Direct calculation gives

$$r_i - 1 = \frac{L - 1}{h_i}, \qquad r_i + \frac{k_{i-1}}{r_{i-1}} = 1 + k_{i-1}, \qquad \prod_i r_i = L.$$

The transformation leaves u_i, v_i unchanged and sends z_i to $z_i' = r_iz_i$. The second identity above proves (3) is unchanged. Therefore $K' = K$ and $F' = F$. When $L \neq 1$, the first identity shows that the two networks are distinct. Furthermore,

$$\alpha' = \prod_i a_i' = \frac{1}{\delta}, \qquad \delta' = \prod_i d_i' = \frac{1}{\alpha},$$

so

$$\det(I_6 - G') = (1 - 1/\delta)(1 - 1/\alpha) \neq 0$$

whenever $\alpha\delta \neq 0$ and the original network is well-posed. The two solutions exist on a nonempty Zariski open set, and the quadratic bound proves they exhaust the generic fiber. The Jacobian at the second point is also nonsingular since $\alpha'\delta' = 1/(\alpha\delta) \neq 1$. □

A small exact numerical example. Let

$$U = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

The following two networks have the same zero pattern:

$$G_1 = \begin{pmatrix} \frac{1}{2}U & 0 \\ I_3 & \frac{1}{3}U^T \end{pmatrix}, \qquad G_2 = \begin{pmatrix} 3U & 0 \\ 6I_3 & 2U^T \end{pmatrix}.$$

They satisfy

$$C(I_6 - G_1)^{-1}B = C(I_6 - G_2)^{-1}B = \frac{1}{91}\begin{pmatrix} 129 & 69 & 75 \\ 75 & 129 & 69 \\ 69 & 75 & 129 \end{pmatrix},$$

and

$$\det(I_6 - G_1) = \frac{91}{108}, \qquad \det(I_6 - G_2) = 182.$$

Their equality is immediate from

$$F^{-1} = \frac{7}{6}I_3 - \frac{1}{2}U - \frac{1}{3}U^T,$$

using $UU^T = I_3$. The numerical coincidences within groups of three edges are only a convenient specialization. Proposition 1 treats all nine edges independently and proves generic ambiguity; the phenomenon does not depend on those coincidences. The verification script also checks a rational example with nine distinct edge values.

Dynamic stability versus a spectral-radius restriction. Both numerical examples above are proper, causal, stable constant transfer matrices. Their internal closed-loop transfer matrices are also constant and well-posed, hence stable in the usual input-output sense. There is no state recursion $w_{t+1} = Gw_t$ here: the node relation is $w_t = Gw_t + Br_t$.

A nonconstant stable rational specialization is also available. Let

$$a(q) = \frac{q}{2(q+1/4)}, \qquad d(q) = \frac{q}{3(q+1/5)}, \qquad A = a(q)U, \quad D = d(q)U^T, \quad X = I_3.$$

The companion network has

$$A' = d(q)^{-1}U, \qquad D' = a(q)^{-1}U^T, \qquad X' = [a(q)d(q)]^{-1}I_3.$$

All entries are proper and stable. Original closed-loop cycle poles solve $a(q)^3 = 1$ or $d(q)^3 = 1$. For a cube root of unity ω, the respective solutions have magnitudes at most

$$\left|\frac{2\omega(1/4)}{1-2\omega}\right| \le \frac{1}{2}, \qquad \left|\frac{3\omega(1/5)}{1-3\omega}\right| \le \frac{3}{10}.$$

The alternative diagonal closed-loop blocks have the same two sets of cycle poles, exchanged; the off-diagonal block equals the original F. Thus both complete networks have stable proper internal closed-loop transfer matrices, as well as stable proper edge transfer functions.

Nevertheless, if the admissible set is additionally restricted by $\rho(G) < 1$ at the frequency under consideration, this counterexample does not apply. Indeed, the block triangular structure gives

$$\rho(G) = \max\{|\alpha|^{1/3}, |\delta|^{1/3}\}.$$

If $|\alpha|, |\delta| < 1$, the alternative has $\alpha' = 1/\delta$ and $\delta' = 1/\alpha$, and violates that restriction. The 2024 manuscript explicitly adds a spectral-radius assumption [2]; it should not be conflated with ordinary stability of rational transfer matrices. A version imposing strictly proper edges likewise requires separate analysis.

Context and verification. The 2020 reference poses the local-to-global question in Section II and its conclusion [1]. Section 12.2 of Legat's 2025 thesis still discusses it as open and reports a search with up to five unknown edges under stated exclusions [3]. The present construction has nine. No claim is made here about minimality or priority in the literature.

The accompanying `verify_counterexample.py` uses exact SymPy arithmetic to verify the rational identity $K' = K$ in nine independent symbols, the Jacobian determinant, the quadratic fiber equation, and two rational numerical pairs. It independently constructs the full 6×6 resolvents and 9×9 observation Jacobians at both points of each pair. Results are recorded in `verification_results.json`.

References

[1] A. Legat and J. M. Hendrickx, "Local Network Identifiability with Partial Excitation and Measurement," *59th IEEE Conference on Decision and Control*, 2020, pp. 4342–4347. Author's PDF.

[2] A. Legat and J. M. Hendrickx, "Identifiability in Networked Systems: Algebraic and Combinatorial Conditions," manuscript listed by the author as submitted in August 2024. Author's PDF.

[3] A. Legat, *Generic local identifiability in dynamical networks: from algebraic characterization to algorithms for signal selection*, Ph.D. thesis, UCLouvain, 2025, Section 12.2, pp. 162–164. Repository record.